

Intro to hyperparameter importance

Hyperparameter optimisation lab: Lecture 02

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8th Apr, 2026



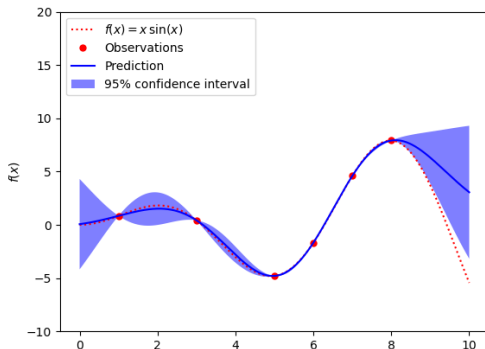
Surrogate model

- ▶ Traditionally, Gaussian processes:

Statistical method that models the data as a multivariate normal distribution

~> Not good with discrete inputs (integer and categorical hyperparameters)

~> Doesn't scale well

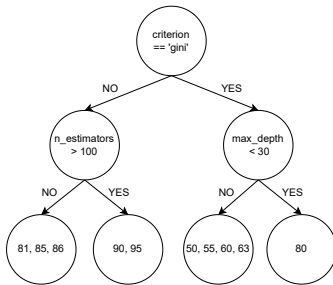


Surrogate model

- ▶ Random forest:
 - ▶ Variance calculated based on predictions of the trees:
 - ▶ Each tree predicts a value and computing the variance as the variance of the trees' predictions

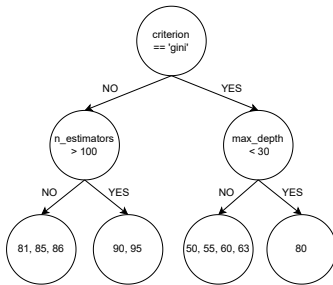
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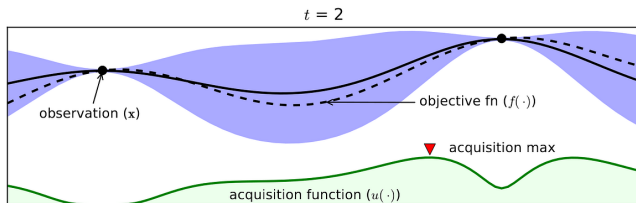
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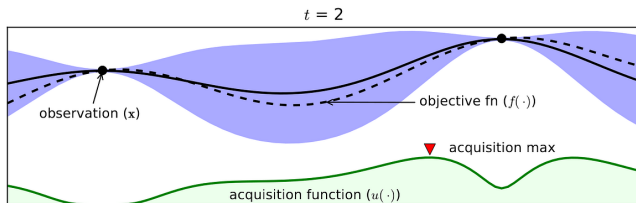
- ▶ Every other ML model that can return mean and variance

Local parameter importance



Key Idea: Use trained surrogate model

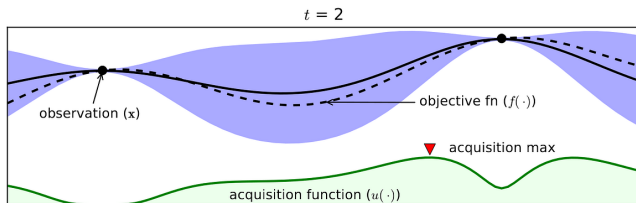
Local parameter importance



Key Idea: Use trained surrogate model

$$VAR_{\lambda}(i) = \sum_{v \in \lambda_i} (\mathbb{E}[\mathcal{L}](\lambda) - \mathcal{L}(\lambda[\lambda_i := v]))^2$$

Local parameter importance



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$$VAR_{\lambda}(i) = \sum_{v \in \lambda_i} (\mathbb{E}[\mathcal{L}](\lambda) - \mathcal{L}(\lambda[\lambda_i := v]))^2$$

$$LPI(i|\lambda) = \frac{VAR_{\lambda}(i)}{\sum_j VAR_{\lambda}(j)}$$

Ablation study

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- ▶ From default to automatically optimized configuration, which steps are important?

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- ▶ Singular effects: Assess λ^{start} with all hyperparameter values from λ^{end}

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- ▶ Singular effects: Assess λ^{start} with all hyperparameter values from λ^{end}
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- ▶ Tradeoff: Find a greedy way from λ^{start} to λ^{end}

Ablation study

- ▶ Given

$$\lambda^{start} = [1, 1, 0, 100], \mathcal{L} = 20\%$$

$$\lambda^{end} = [0.98, 2.42, 1, 42], \mathcal{L} = 4\%$$

$$\lambda^1 = [0.98, 1, 0, 100], \mathcal{L} = 19\%$$

$$\lambda^2 = [1, 2.42, 0, 100], \mathcal{L} = 16\%$$

$$\lambda^3 = [1, 1, 1, 100], \mathcal{L} = 7\%$$

$$\lambda^4 = [1, 1, 0, 42], \mathcal{L} = 12\%$$

- ▶ 1st step: Flip hyperparameter three
- ▶ Continue with $\lambda^1 = [1, 1, 1, 100]$

Full ablation path

$$\lambda^{start} = [1, 1, 0, 100], \mathcal{L} = 20\%$$

$$\lambda^1 = [1, 1, 1, 100], \mathcal{L} = 7\%$$

$$\lambda^2 = [0.98, 1, 1, 100], \mathcal{L} = 5\%$$

$$\lambda^3 = [0.98, 2.42, 1, 100], \mathcal{L} = 4\%$$

$$\lambda^4 = [0.98, 2.42, 1, 42], \mathcal{L} = 4\%$$

Global importance analysis

- ▶ What is the importance of a hyperparameter by marginalising over all other hyperparameter effects?
- ▶ Use surrogate model to compute these effects

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Definition (fANOVA)

Write performance predictions as sum of components:

$$\hat{y}(\lambda_1, \dots, \lambda_n) = \hat{f}_0 + \sum_{i=1}^n \hat{f}_i(\lambda_i) + \sum_{i \neq j} \hat{f}_{ij}(\lambda_i \lambda_j) + \dots$$

$\hat{y}(\lambda_1, \dots, \lambda_n)$ = average response + main effects +
2-D interaction effects + higher order effects

fANOVA analysis

Definition (Variance decomposition)

Calculate contribution the total variance V :

$$V = \frac{1}{||\Lambda||} \int_{\lambda_1} \int_{\lambda_n} [(\hat{y}(\lambda) - \hat{f}_0)^2] d\lambda_1, \dots d\lambda_n$$

Theorem

The fANOVA and variance decomposition can be done efficiently in linear time if the surrogate model is a random forest. Hutter et al. 2014

fANOVA analysis

- ▶ How much variance can be explained by a hyperparameter (or combinations of hyperparameters) marginalized over all other parameters?

Hyperparameter	Explained Variance
Discount rate	19.3 %
Batch size	15.7 %
Learning rate	3.7 %
Likelihood ration clipping	3.4%
...	
discount rate & batch size	10.4%
discount rate & likelihood ration clipping	4.4%
...	

Exemplary analysis of PPO on cartpole

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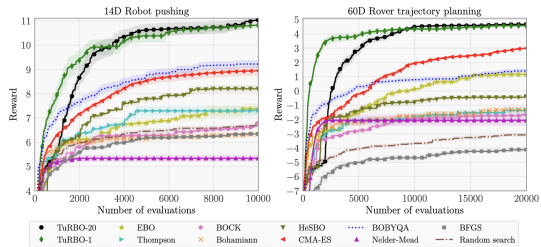
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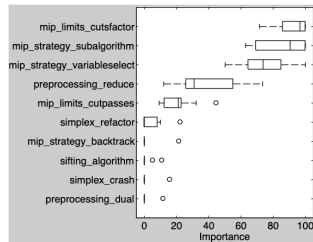
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- ▶ Use convergence plots with confidence intervals for the optimisation process
- ▶ Use box plots to show variance in results

Experiment setup: Visualization



Convergence Plot



Box Plot

DeepCAVE

- ▶ Tool for analysis and visualisation of HPO
- ▶ Contains fANOVA, LPI, and other analysis



DeepCAVE

Assignment 1

- ▶ Implementing random search, successive halving, Bayesian optimisation
- ▶ Experimenting with them & visualising the results
- ▶ Deadline: 5 May 2026, 23:59 CEST
- ▶ Meeting on 6 May 2026 for discussion
- ▶ Check Moodle for exact details
- ▶ Bid on papers for 2nd assignment by 5 May 2026, 23:59 CEST

Assignment 1

- ▶ Questions?
- ▶ Remember: Turn in your preferences for the second assignment